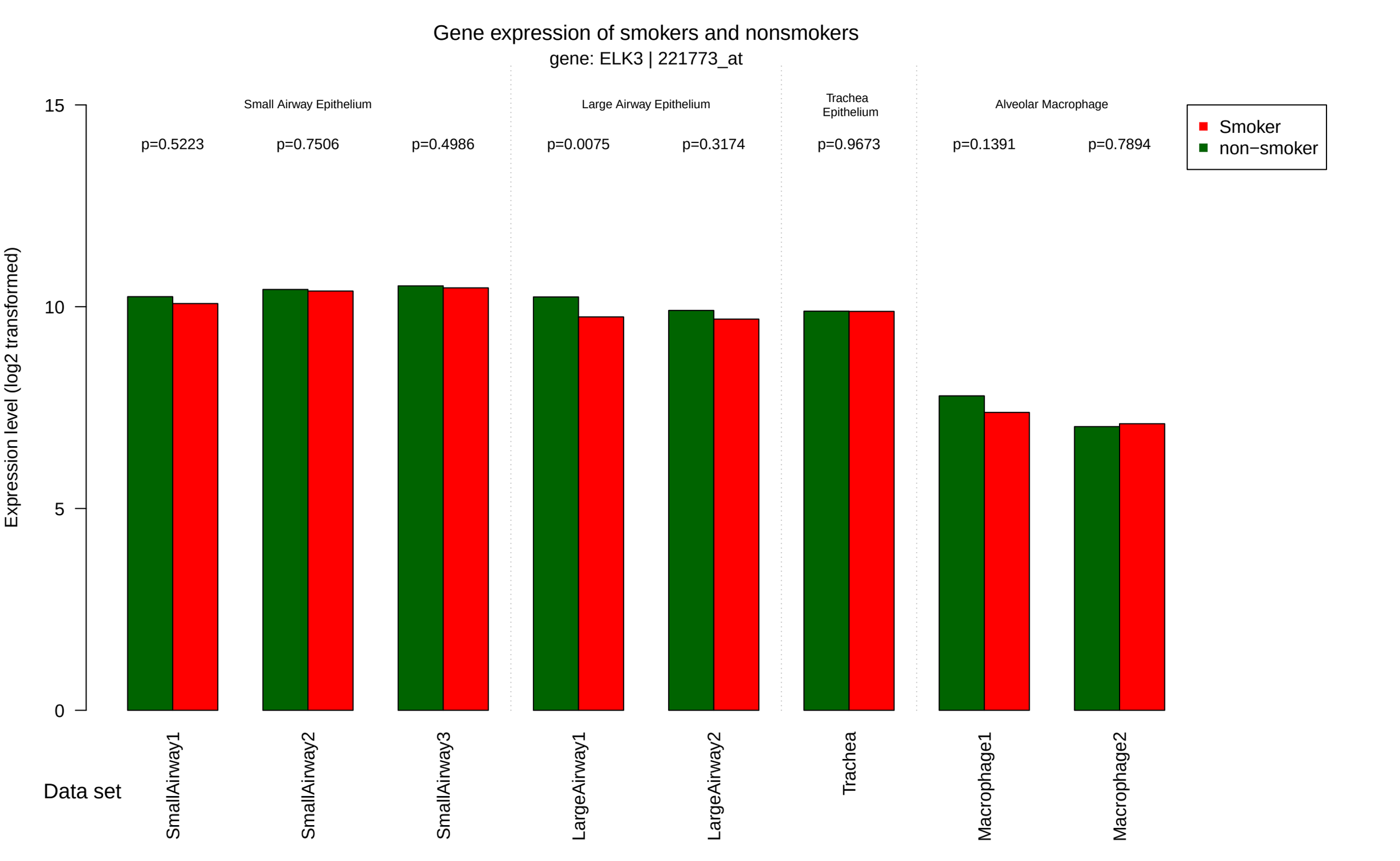
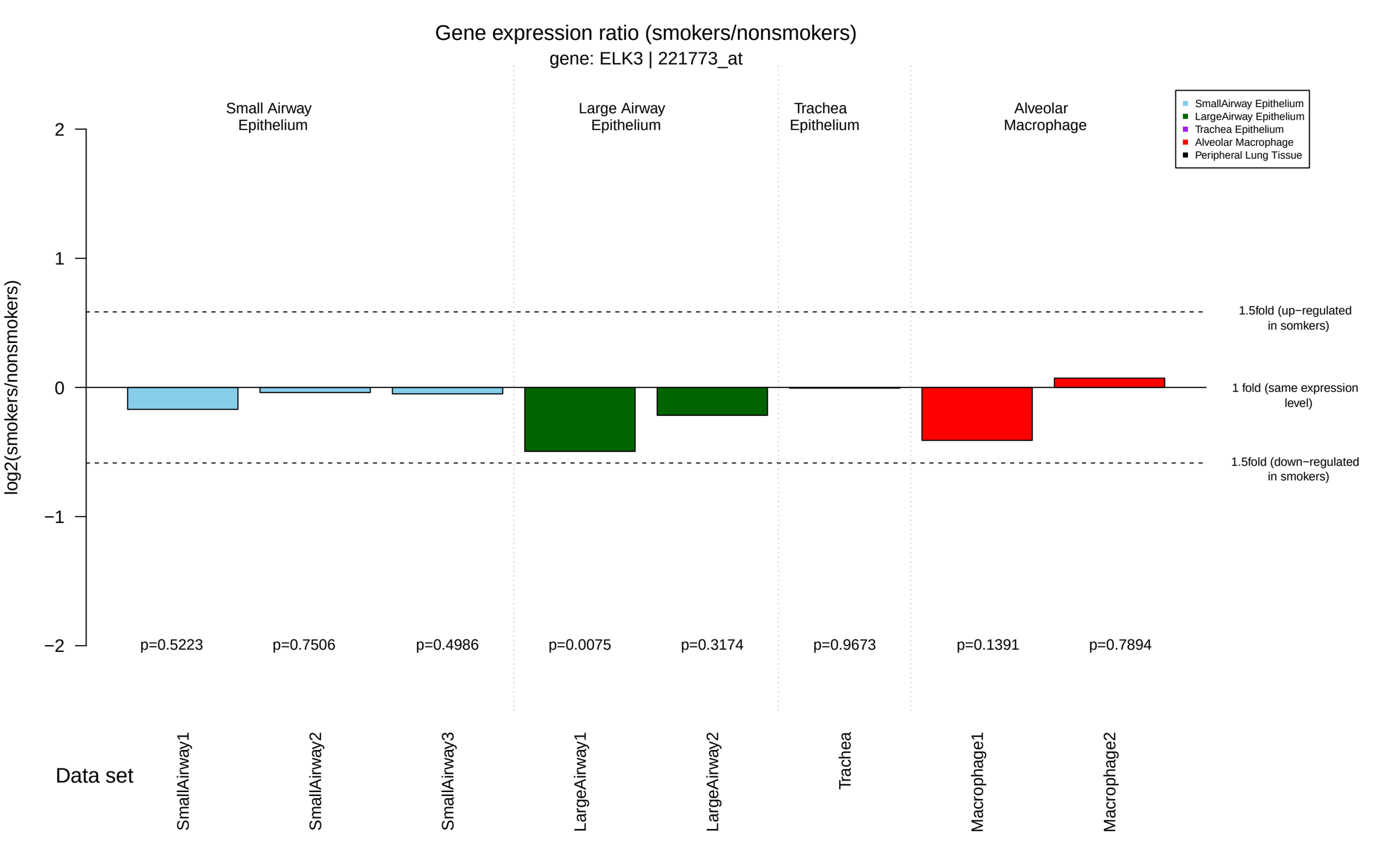
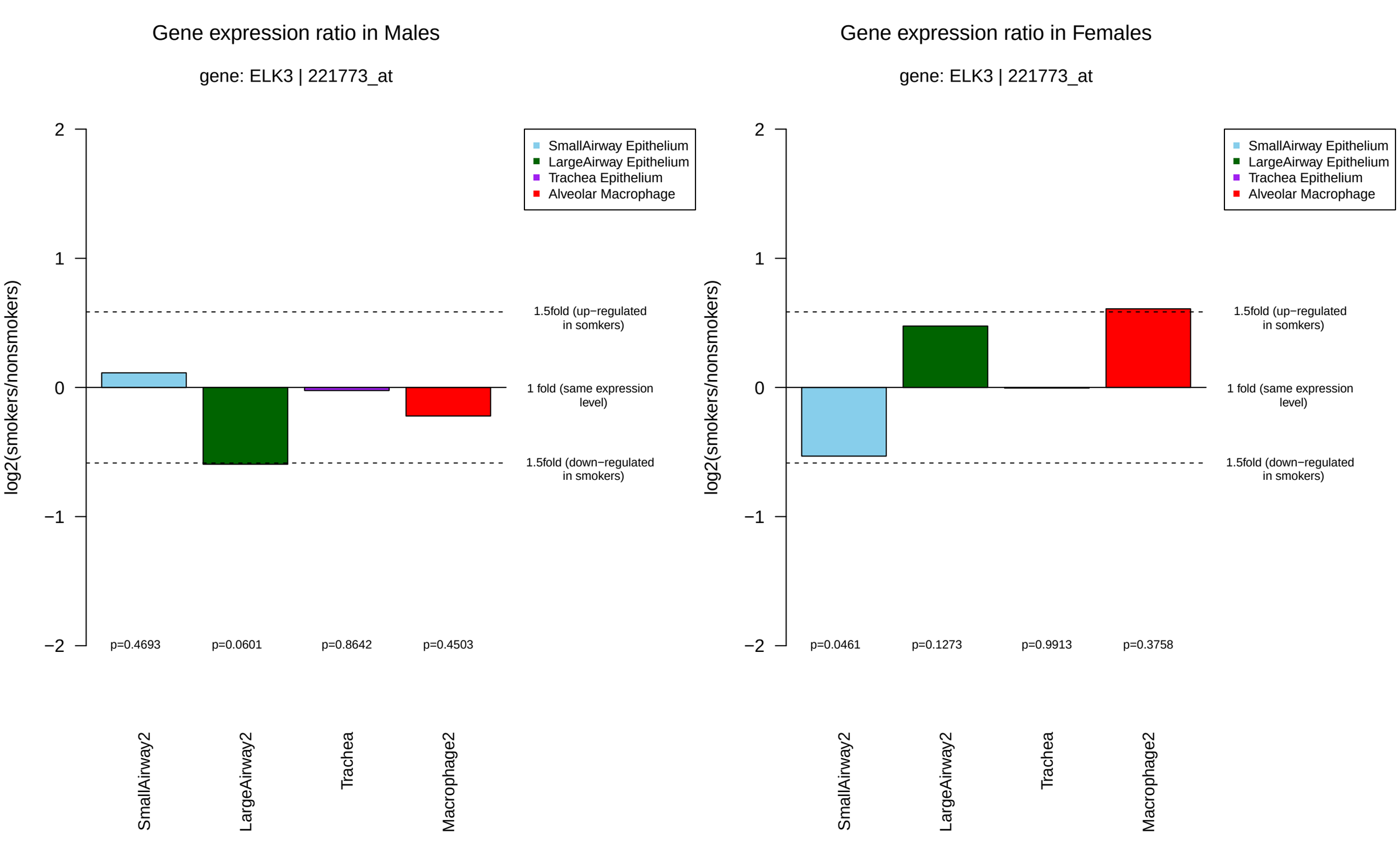
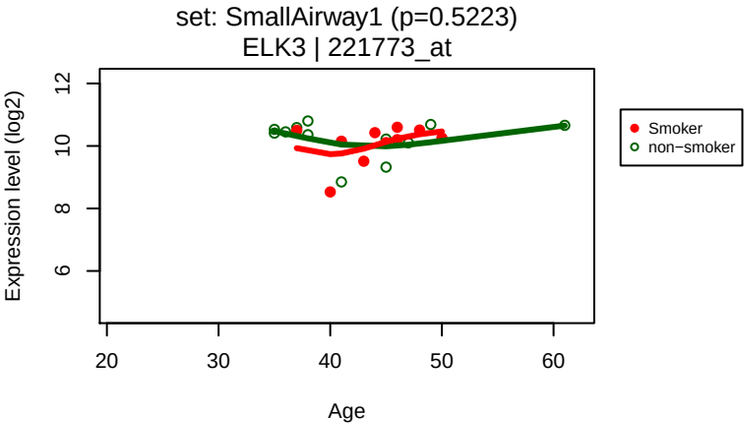
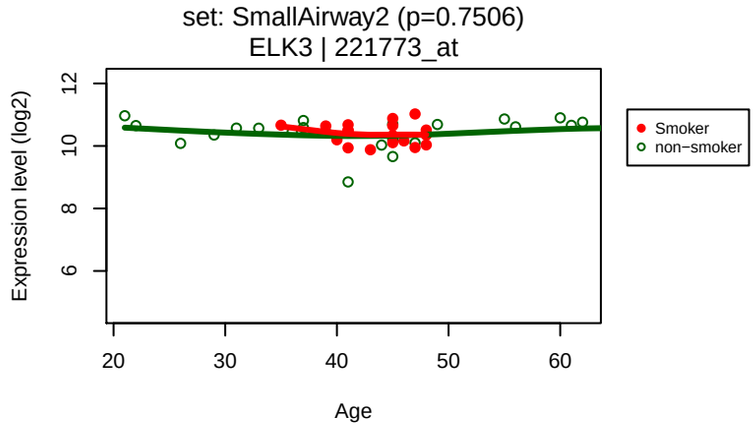
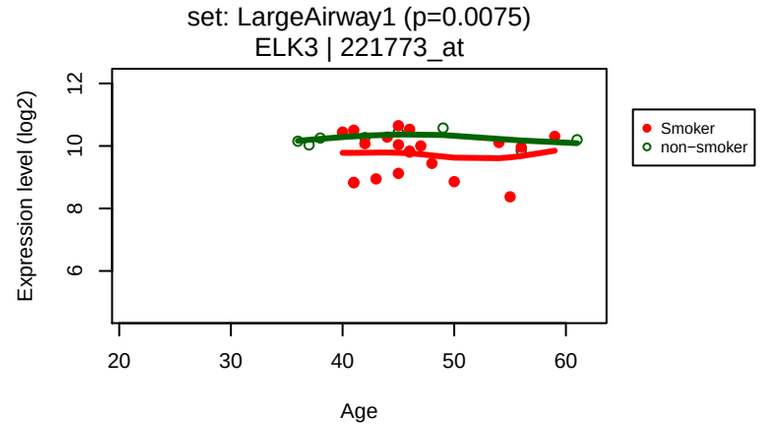

Gene expression of smokers and nonsmokers
gene: ELK3 | 221773_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5223
p=0.7506
p=0.4986
p=0.0075
p=0.3174
p=0.9673
p=0.1391
p=0.7894
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

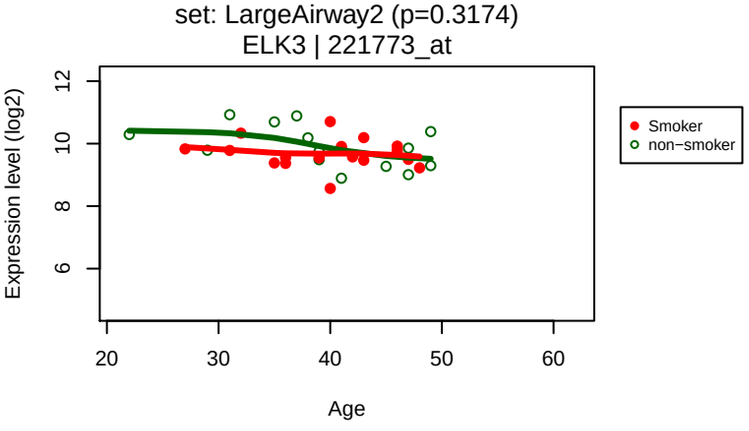

Gene expression ratio (smokers/nonsmokers)
gene: ELK3 | 221773_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5223
p=0.7506
p=0.4986
p=0.0075
p=0.3174
p=0.9673
p=0.1391
p=0.7894
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

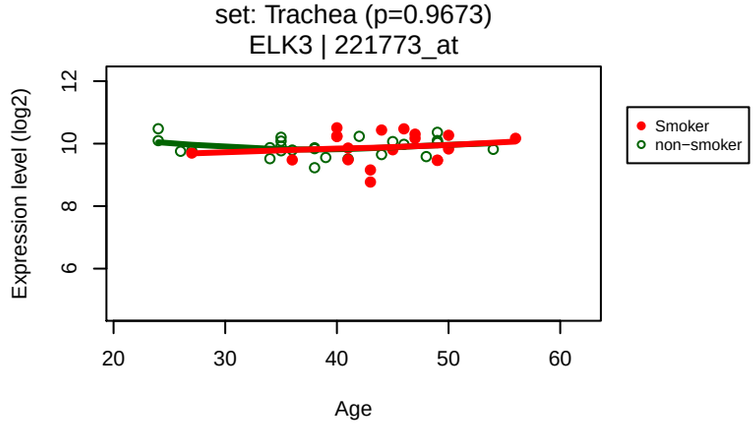

Gene expression ratio in Males
gene: ELK3 | 221773_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4693
p=0.0601
p=0.8642
p=0.4503
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ELK3 | 221773_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0461
p=0.1273
p=0.9913
p=0.3758
SmallAirway2
LargeAirway2
Trachea
Macrophage2

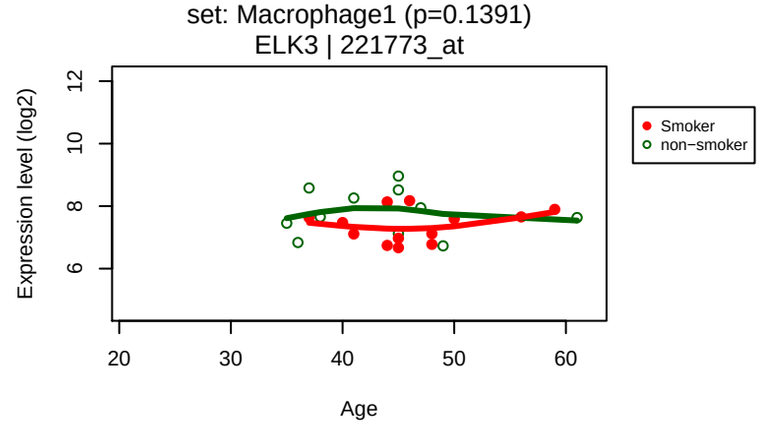

set: SmallAirway1 (p=0.5223)
ELK3 | 221773_at
Expression level (log2)
Age
Smoker
non−smoker

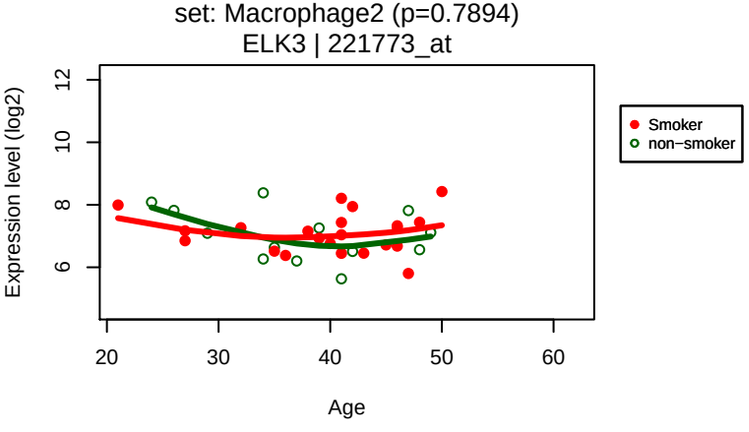

set: SmallAirway2 (p=0.7506)
ELK3 | 221773_at
Expression level (log2)
Age
Smoker
non−smoker

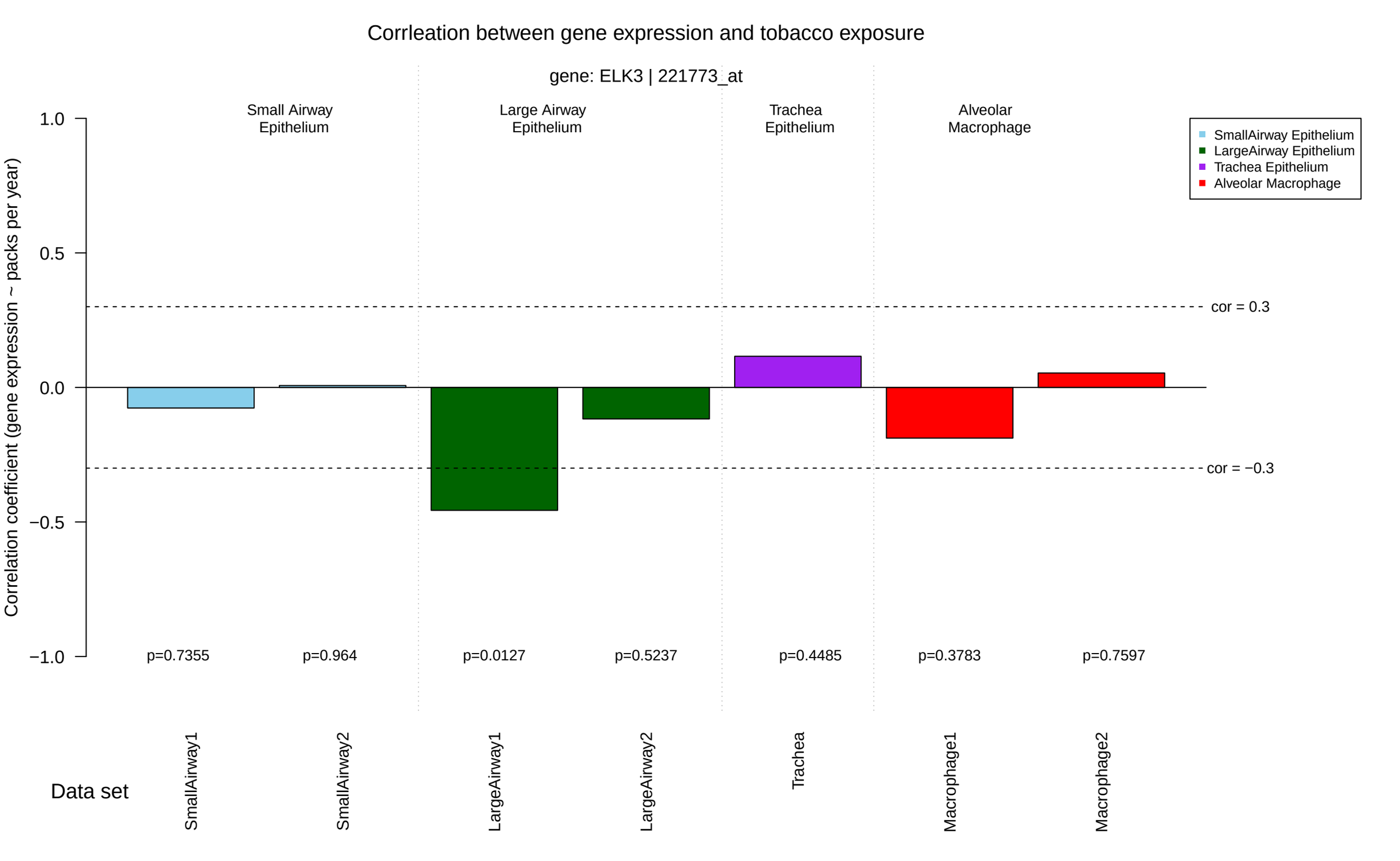

set: LargeAirway1 (p=0.0075)
ELK3 | 221773_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3174)
ELK3 | 221773_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9673)
ELK3 | 221773_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1391)
ELK3 | 221773_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7894)
ELK3 | 221773_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ELK3 | 221773_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7355
p=0.964
p=0.0127
p=0.5237
p=0.4485
p=0.3783
p=0.7597
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2